

Probability and Statistics (MT-2005)

Homework # 03(a)

Random Variables, PMF, PDF, and CDF

1. A factory produces light bulbs, and the number of defective bulbs in a batch of 5 is modeled as a discrete random variable X with the following probability mass function:

$$P(X = x) = \begin{cases} 0.3, & x = 0 \\ 0.4, & x = 1 \\ 0.2, & x = 2 \\ 0.1, & x = 3 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Verify that this is a valid PMF.
 - (b) Compute the cumulative distribution function (CDF), $F(X)$.
 - (c) Compute Mean, Median, Mode, Quartiles(Q_1, Q_3), Geometric mean and Harmonic mean of random variable X .
 - (d) Compute $Var(X)$.
2. Let X be a continuous random variable with the probability density function:

$$f(x) = \begin{cases} kx^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Find k such that $f(x)$ is a valid PDF.
 - (b) Compute the cumulative distribution function (CDF), $F(x)$.
 - (c) Compute Mean, Median, Mode, Quartiles(Q_1, Q_3), Geometric mean and Harmonic mean of random variable X .
 - (d) Compute $Var(X)$.
 - (e) Find $P(0.2 \leq X \leq 0.8)$.
 - (f) Find $P(|X| \leq 3/2)$.
 - (g) Find $P(X \leq .7 | 0.2 \leq X \leq 0.8)$.
3. A continuous random variable X has the CDF:

$$F(x) = \begin{cases} 0.2 & \text{for } x \leq 1 \\ 0.7 & \text{for } 1 < x \leq 3 \\ 1 & \text{for } x > 3 \end{cases}$$

- (a) Find PDF.
- (b) What is the probability that X takes a value between 1 and 3, i.e., $P(1 < X \leq 3)$?